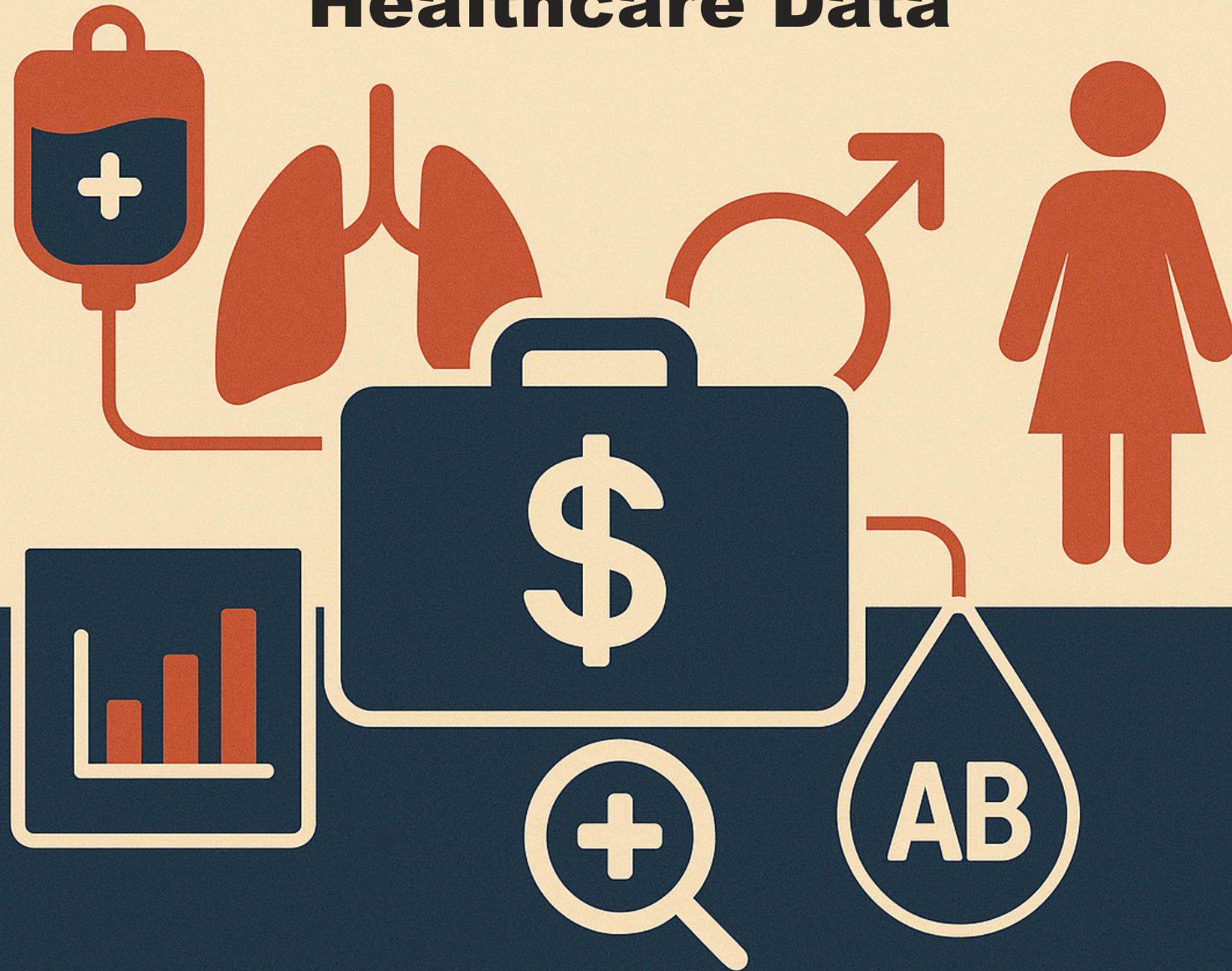


"Analyzing Cost Drivers and Risk Factors in Healthcare Data"



Problem statements:

1) Which medical conditions drive the highest costs, longest stays, and worst test outcomes, and how can we improve treatment efficiency for them?

2) Are there any specific age groups, genders, or blood groups that are more frequently associated with high-cost conditions?



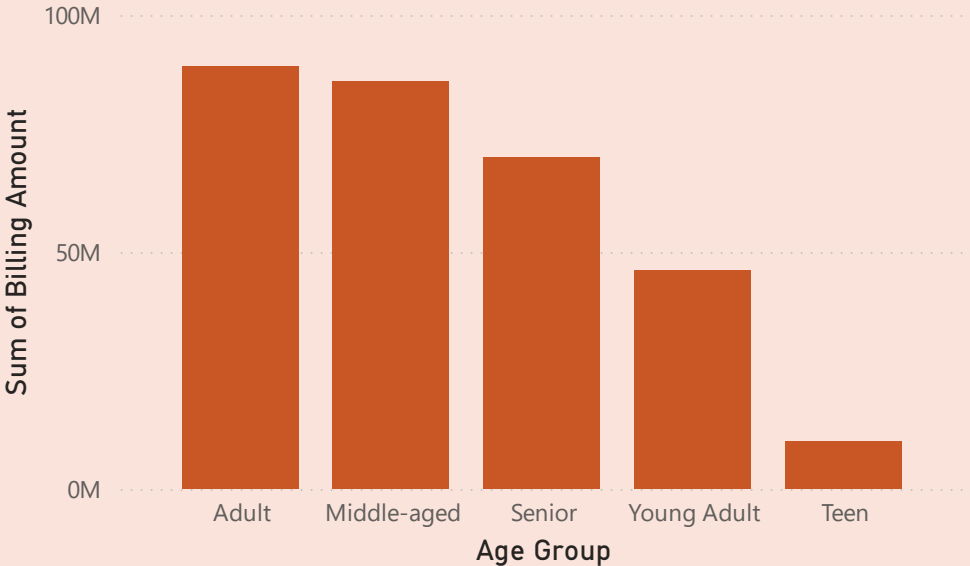
Cost Analysis by Demographics

Insights:

- 1) Worry about high medical bill if u have A-, AB- or A+ blood group.
- 2) Adult and Middle-aged pay more in medical bills then Old people.
- 3) Bare minimum difference between male/female test results to average bill amount.
- 4) Blood groups mentioned in '1' are mostly dominant in all medical conditions other than diabetes where B+ dominates.

➔ Analysis for Highly Billed Age Groups

Sum of Billing Amount by Age Group



Avg. Stay in Days

121.47

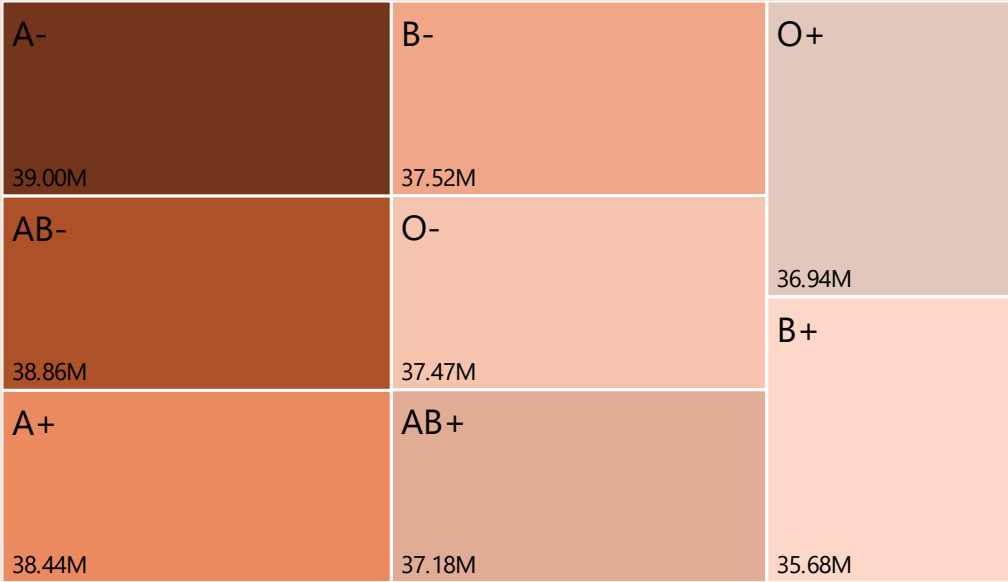
Avg. of Age

51.66

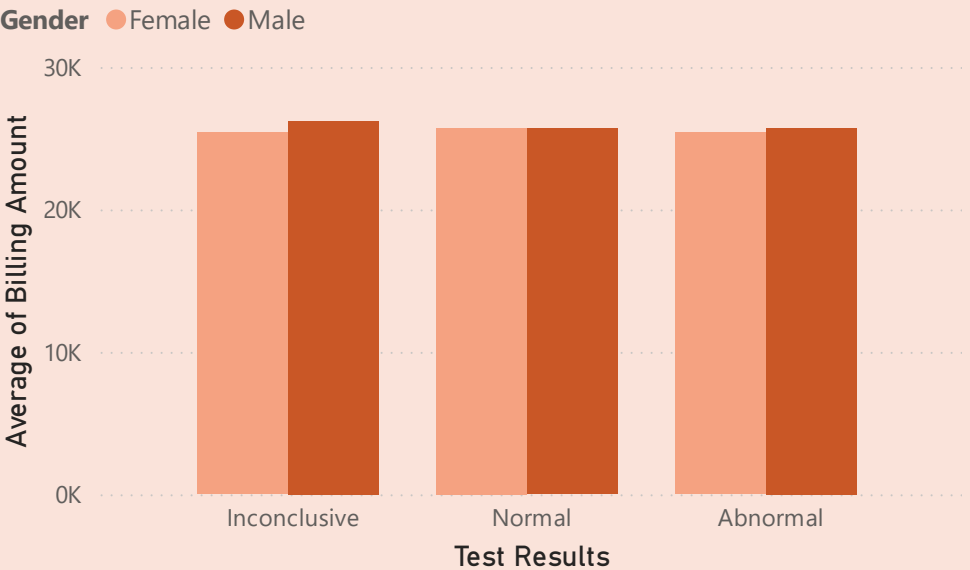
Total Billing Amount

301.09M

Sum of Billing Amount by Blood Type



Average of Billing Amount by Test Results and Gender



Select Medical Condition

Arthritis

Asthma

Cancer

Diabetes

Hypertension

Obesity

Analysis of Highest Charged Age Groups

Insurance Provider

Aetna

Blue Cross

Cigna

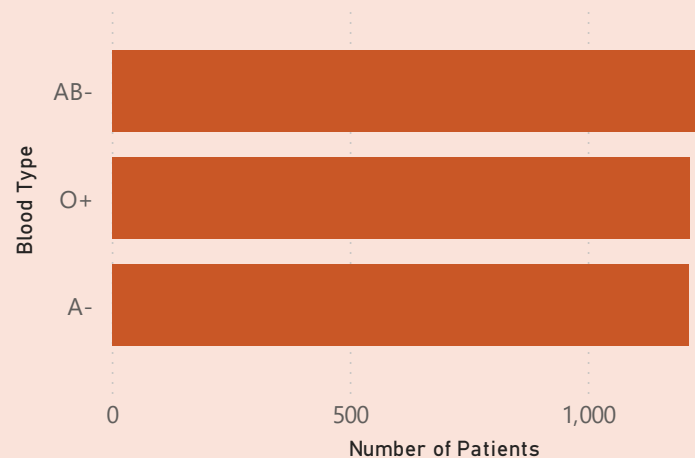
Medicare

UnitedHealthcare

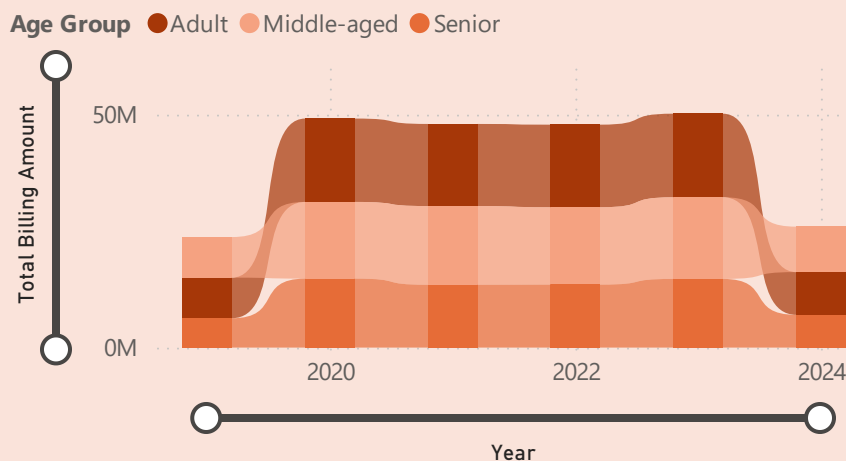
Analysis of Common Blood Groups in Highly Billed Age Groups



Most Common Blood Types



Total Billing Amount by Year and Age Group



Insights:

• Blood Type Frequency:

The three most common blood groups within highly billed age groups are AB-, O+, and A-.

• Distribution of Admission Types:

The three admission types have a nearly equal distribution of admissions.

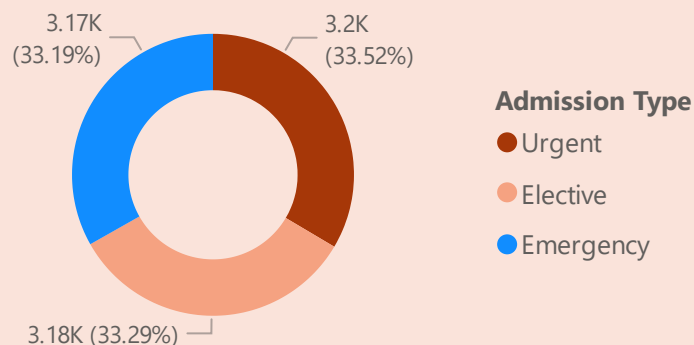
• Historical Billing Amount Trend:

Adults accounted for the highest total billing amount between 2020 and 2023.

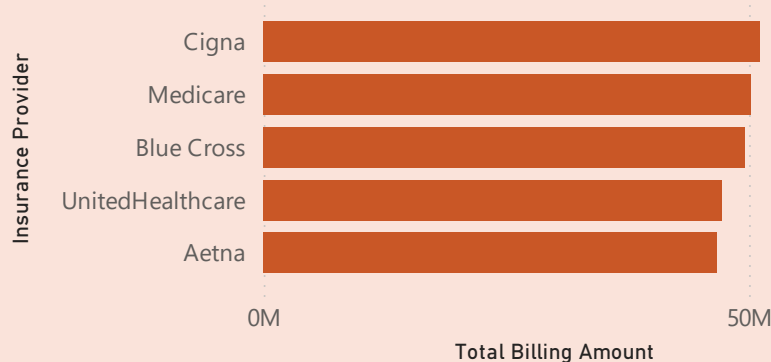
• Insurance Provider Trends:

Cigna has the highest total billing amount

Number of Patients per Admission Type



Total Billing Amount by Insurance Provider



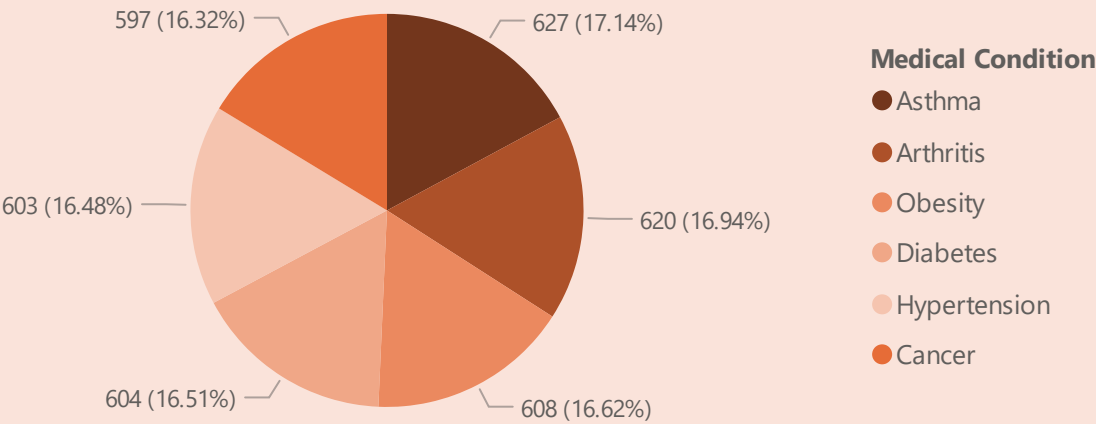
Gender

Female

Male

ANALYSIS BY TOP AGE GROUPS AND BLOOD TYPES:

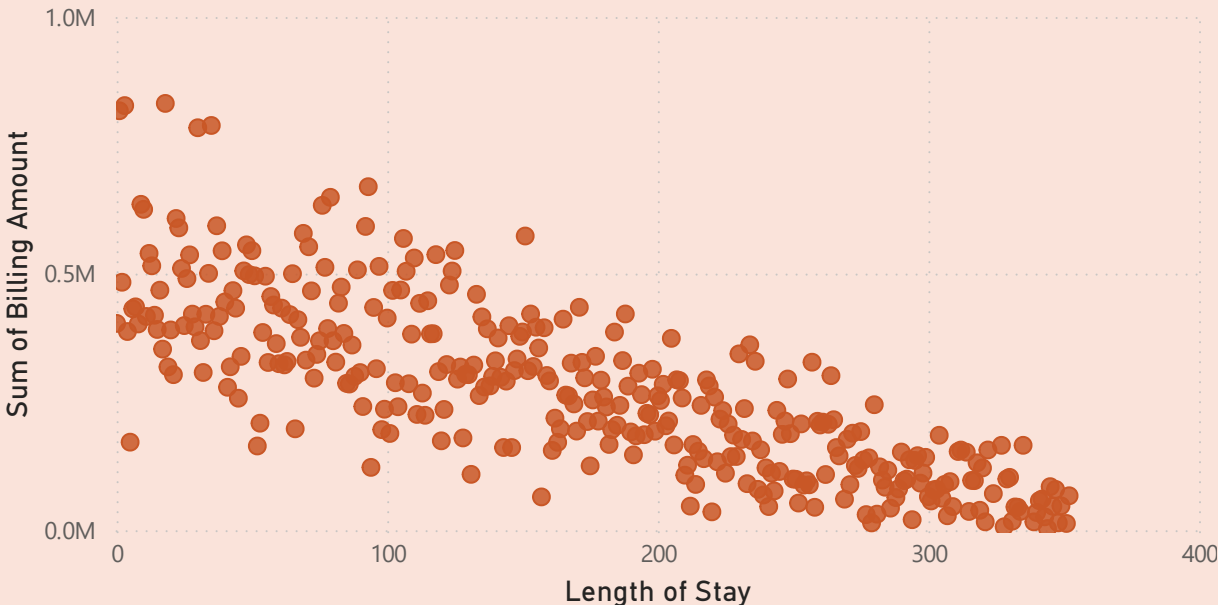
Distribution of Patients across Different Medical Conditions



Billing Trend Over Time



Relation between Billing Amount & Length of Stay



Key Insights:

- **Top Conditions by Volume:**
Asthma and Arthritis account for the highest number of cases.
- **Seasonal Billing Trends:**
Billing peaks recur every Q1 and Q3, indicating seasonal spikes.
- **Length of Stay Impact:**
Billing drops with longer hospital stays - strong negative correlation.

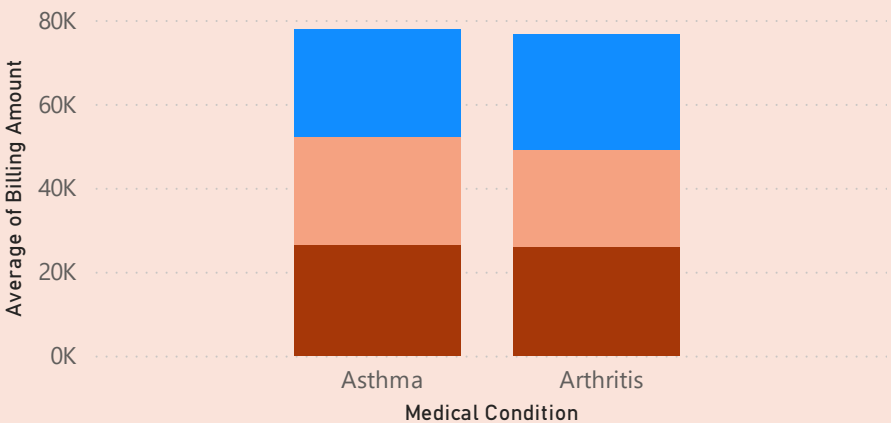
Medical Condition Analysis for Asthma & Arthritis Patients:



MEDICAL CONDITION ANALYSIS:

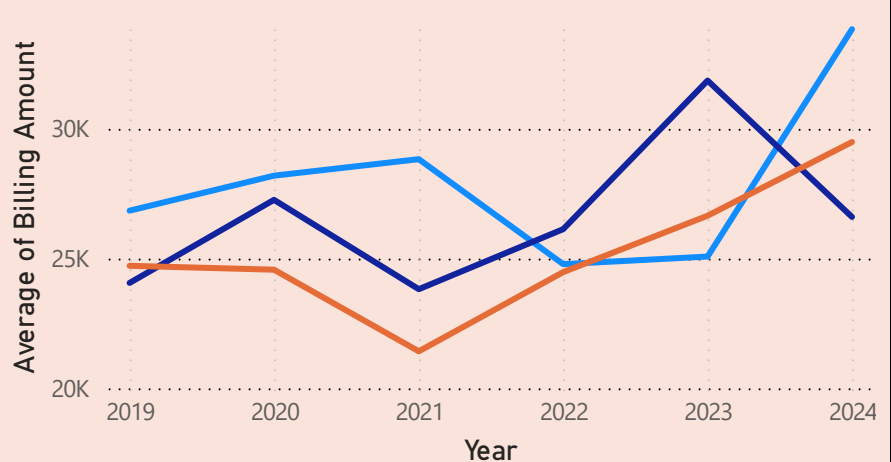
Average of Billing Amount by Medical Condition and Age Group

Age Group ● Adult ● Middle-aged ● Senior



Average of Billing Amount by Year and Medication

Medication ● Aspirin ● Ibuprofen ● Paracetamol

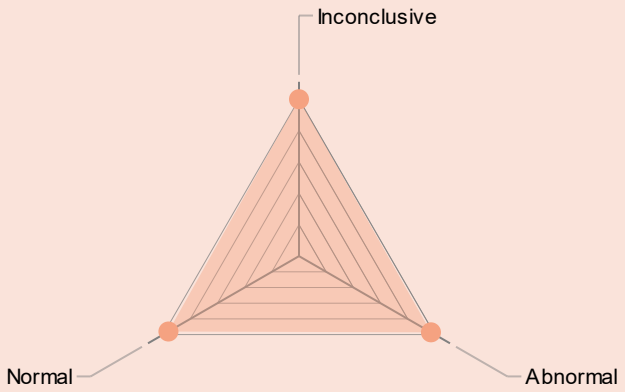


Insights:

- **Age group trends:**
Asthma shows the highest billing among adults, while Arthritis peaks in seniors.
- **Billing Trends by Medication:**
Billing amounts for Aspirin and Ibuprofen fluctuate with time, while Paracetamol shows a steadier rise.
- **Test Results Impact:**
Inconclusive test results tend to have higher average billing compared to normal and abnormal results.
- **Insurance Provider Trends:**
Cigna handles the highest volume of billing cases for both Asthma and Arthritis, followed by UnitedHealthcare and Blue Cross.

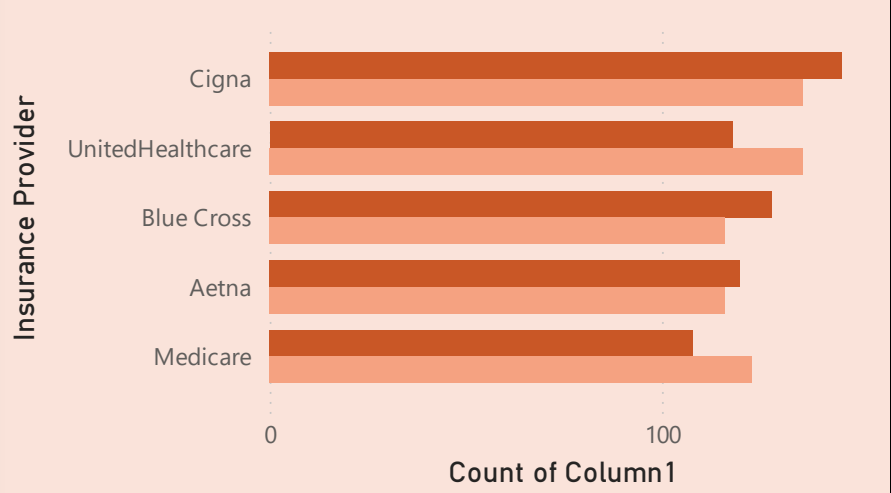
Average of Billing Amount by Test Results

Axis ● Average of Billing Amount



Number of Patients per Insurance Provider and Medical Condition

Medical Condition ● Arthritis ● Asthma



Final Conclusions



CONCLUSION:

Cost by Demographics:

Middle-aged and adult patients contribute the most to total billing, highlighting them as the most resource-intensive age groups.

Top 3 Age Groups:

Among the top 3 age groups, adults incur the highest billing.

Age and Blood Group:

Seniors with blood types AB- and O+ incur higher billing, especially for chronic conditions like arthritis and asthma.

Medical Conditions:

Chronic conditions such as arthritis and asthma are most common, with inconclusive test results leading to the highest treatment costs.